Appendix C

The complete derivation steps for transforming the bi-level DSO-SCHO[1] interaction problem in (16) into an equivalent MILP problem are as follows.

KKT conditions derivation of the DSO problem

Here, the complete primal problem of the DSO problem is expressed as follows:

$$
\begin{aligned}
&\min_{\{P_{\mathrm{sub},t}^{\phi},P_{j,t}^{\mathrm{MT},\phi}\}} f_{\mathrm{DSO}} = \sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\lambda_t^{\phi}P_{\mathrm{sub},t}^{\phi} + \sum_{j\in\Omega_{\mathrm{MT}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}c_j^{\mathrm{MT}}P_{j,t}^{\mathrm{MT},\phi}\\
&s.t.\\
&P_{\mathrm{sub},t}^{\phi} = \sum_{k\in C_1}P_{1k,t}^{\phi} : \pi_{1,t}^{p,\phi}, \forall t\in\Omega_{\mathrm{T}}, \forall\phi\in\Phi\\
&Q_{\mathrm{sub},t}^{\phi} = \sum_{k\in C_1}Q_{1k,t}^{\phi} : \pi_{1,t}^{q,\phi}, \forall t\in\Omega_{\mathrm{T}}, \forall\phi\in\Phi\\
&\sum_{i\in A_j}P_{ij,t}^{\phi} + P_{j,t}^{\mathrm{MT},\phi} - P_{j,t}^{\mathrm{D},\phi} - P_{j,t}^{\mathrm{Hub},\phi} = \sum_{k:j\to k}P_{jk,t}^{\phi} : \pi_{j,t}^{p,\phi}, \forall j\in\Omega_{\mathrm{N}}, \forall t\in\Omega_{\mathrm{T}}, \forall\phi\in\Phi\\
&\sum_{i\in A_j}Q_{ij,t}^{\phi} + Q_{j,t}^{\mathrm{MT},\phi} - Q_{j,t}^{\mathrm{D},\phi} = \sum_{k:j\to k}Q_{jk,t}^{\phi} : \pi_{j,t}^{q,\phi}, \forall j\in\Omega_{\mathrm{N}}, \forall t\in\Omega_{\mathrm{T}}, \forall\phi\in\Phi\\
&v_{j,t}^{\phi} = v_{i,t}^{\phi} - 2\left(\tilde{\boldsymbol{r}}_{ij}^{\phi}\boldsymbol{P}_{ij,t}^{\Phi} + \tilde{\boldsymbol{x}}_{ij}^{\phi}\boldsymbol{Q}_{ij,t}^{\Phi}\right) : \theta_{j,t}^{v,\phi}, \forall j\in\Omega_{\mathrm{N}}, \forall t\in\Omega_{\mathrm{T}}, \forall\phi\in\Phi\\
&\left(V_j^{\min,\phi}\right)^2 \le v_{j,t}^{\phi} \le \left(V_j^{\max,\phi}\right)^2 : \gamma_{j,t}^{v,\phi,\min}, \gamma_{j,t}^{v,\phi,\max}, \forall j\in\Omega_{\mathrm{N}}, \forall t\in\Omega_{\mathrm{T}}, \forall\phi\in\Phi\\
&v_{1,t}^{\phi} = \left(V^{ref}\right)^2 : \theta_{1,t}^{v,\phi}, \forall t\in\Omega_{\mathrm{T}}, \forall\phi\in\Phi\\
&-P_{\max} \le P_{\mathrm{sub},t}^{\phi} \le P_{\max} : \gamma_{1,t}^{p,\phi,\min}, \gamma_{1,t}^{p,\phi,\max}, \forall t\in\Omega_{\mathrm{T}}, \forall\phi\in\Phi\\
&-Q_{\max} \le Q_{\mathrm{sub},t}^{\phi} \le Q_{\max} : \gamma_{1,t}^{q,\phi,\min}, \gamma_{1,t}^{q,\phi,\max}, \forall t\in\Omega_{\mathrm{T}}, \forall\phi\in\Phi\\
&0 \le P_{j,t}^{\mathrm{MT},\phi} \le P_j^{\mathrm{MT},\phi,\max} : \gamma_{j,t}^{\mathrm{MT},p,\phi,\min}, \gamma_{j,t}^{\mathrm{MT},p,\phi,\max}, \forall j\in\Omega_{\mathrm{N}}, \forall t\in\Omega_{\mathrm{T}}, \forall\phi\in\Phi\\
&0 \le Q_{j,t}^{\mathrm{MT},\phi} \le P_{j,t}^{\mathrm{MT},\phi}\tan(\arccos(\varphi^{\mathrm{pf}})) : \gamma_{j,t}^{\mathrm{MT},q,\phi,\min}, \gamma_{j,t}^{\mathrm{MT},q,\phi,\max}, \forall j\in\Omega_{\mathrm{N}}, \forall t\in\Omega_{\mathrm{T}}, \forall\phi\in\Phi\\
&-K_v \le v_{j,t}^{\phi} - v_{j,t}^{\varphi} \le K_v : \gamma_{j,t}^{\phi\varphi,\min}, \gamma_{j,t}^{\phi\varphi,\max}, \forall j\in\Omega_{\mathrm{N}}, \forall t\in\Omega_{\mathrm{T}}, \forall(\phi,\varphi)\in\Phi, \phi\ne\varphi
\end{aligned}
\tag{C1}
$$

Then the primal problem with constraints can be converted to the equivalent optimization problem without constraints by Lagrangian relaxation. The Lagrangian function of (C1) is as

$$
\begin{aligned}
&L_{\mathrm{DSO}}\left(\begin{array}{l}
P_{ij,t}^{\phi}, Q_{ij,t}^{\phi}, v_{j,t}^{\phi}, P_{\mathrm{sub},t}^{\phi}, Q_{\mathrm{sub},t}^{\phi}, P_{j,t}^{\mathrm{MT},\phi}, Q_{j,t}^{\mathrm{MT},\phi} \\
\pi_{j,t}^{p,\phi}, \pi_{j,t}^{q,\phi}, \theta_{j,t}^{v,\phi}, \gamma_{j,t}^{v,\phi,\min}, \gamma_{j,t}^{v,\phi,\max}, \gamma_{1,t}^{p,\phi,\min}, \gamma_{1,t}^{p,\phi,\max}, \gamma_{1,t}^{q,\phi,\min}, \gamma_{1,t}^{q,\phi,\max}, \gamma_{j,t}^{\mathrm{MT},p,\phi,\min}, \gamma_{j,t}^{\mathrm{MT},p,\phi,\max}, \\
\gamma_{j,t}^{\mathrm{MT},q,\phi,\min}, \gamma_{j,t}^{\mathrm{MT},q,\phi,\max}, \gamma_{j,t}^{\phi\varphi,\min}, \gamma_{j,t}^{\phi\varphi,\max}
\end{array}\right) \\
&= \sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\lambda_t^{\phi}P_{\mathrm{sub},t}^{\phi} + \sum_{j\in\Omega_{\mathrm{MT}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}c_j^{\mathrm{MT}}P_{j,t}^{\mathrm{MT},\phi} - \sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\pi_{1,t}^{p,\phi}\left(P_{\mathrm{sub},t}^{\phi} - \sum_{k\in C_1}P_{1k,t}^{\phi}\right) - \sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\pi_{1,t}^{q,\phi}\left(Q_{\mathrm{sub},t}^{\phi} - \sum_{k\in C_1}Q_{1k,t}^{\phi}\right) \\
&- \sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\pi_{j,t}^{p,\phi}\left(\sum_{i\in A_j}P_{ij,t}^{\phi} + P_{j,t}^{\mathrm{MT},\phi} - P_{j,t}^{\mathrm{D},\phi} - P_{j,t}^{\mathrm{Hub},\phi} - \sum_{k\in C_j}P_{jk,t}^{\phi}\right) \\
&- \sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\pi_{j,t}^{q,\phi}\left(\sum_{i\in A_j}Q_{ij,t}^{\phi} + Q_{j,t}^{\mathrm{MT},\phi} - Q_{j,t}^{\mathrm{D},\phi} - \sum_{k\in C_j}Q_{jk,t}^{\phi}\right) \\
&- \sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\theta_{j,t}^{v,\phi}\left[v_{j,t}^{\phi} - v_{i,t}^{\phi} + 2\left(\tilde{\boldsymbol{r}}_{ij}^{\phi}\boldsymbol{P}_{ij,t}^{\Phi} + \tilde{\boldsymbol{x}}_{ij}^{\phi}\boldsymbol{Q}_{ij,t}^{\Phi}\right)\right] - \sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\theta_{1,t}^{v,\phi}\left[v_{1,t}^{\phi} - \left(V^{ref}\right)^2\right] \\
&+ \sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\gamma_{j,t}^{v,\phi,\min}\left[\left(V_j^{\min,\phi}\right)^2 - v_{j,t}^{\phi}\right] + \sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\gamma_{j,t}^{v,\phi,\max}\left[v_{j,t}^{\phi} - \left(V_j^{\max,\phi}\right)^2\right] \\
&+ \sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\gamma_{1,t}^{p,\phi,\min}\left(-P_{\max} - P_{\mathrm{sub},t}^{\phi}\right) + \sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\gamma_{1,t}^{p,\phi,\max}\left(P_{\mathrm{sub},t}^{\phi} - P_{\max}\right) \\
&+ \sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\gamma_{1,t}^{q,\phi,\min}\left(-Q_{\max} - Q_{\mathrm{sub},t}^{\phi}\right) + \sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\gamma_{1,t}^{q,\phi,\max}\left(Q_{\mathrm{sub},t}^{\phi} - Q_{\max}\right) \\
&+ \sum_{j\in\Omega_{\mathrm{MT}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\gamma_{j,t}^{\mathrm{MT},p,\phi,\min}\left(-P_{j,t}^{\mathrm{MT},\phi}\right) + \sum_{j\in\Omega_{\mathrm{MT}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\gamma_{j,t}^{\mathrm{MT},p,\phi,\max}\left(P_{j,t}^{\mathrm{MT},\phi} - P_j^{\mathrm{MT},\phi,\max}\right) \\
&+ \sum_{j\in\Omega_{\mathrm{MT}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\gamma_{j,t}^{\mathrm{MT},q,\phi,\min}\left(-Q_{j,t}^{\mathrm{MT},\phi}\right) + \sum_{j\in\Omega_{\mathrm{MT}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\gamma_{j,t}^{\mathrm{MT},q,\phi,\max}\left(Q_{j,t}^{\mathrm{MT},\phi} - P_{j,t}^{\mathrm{MT},\phi}\tan(\arccos(\varphi^{\mathrm{pf}}))\right) \\
&+ \sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\substack{\forall(\phi,\varphi)\in\Phi \\ \phi\neq\varphi}}\gamma_{j,t}^{\phi\varphi,\min}\left(-K_v - v_{j,t}^{\phi} + v_{j,t}^{\varphi}\right) + \sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\substack{\forall(\phi,\varphi)\in\Phi \\ \phi\neq\varphi}}\gamma_{j,t}^{\phi\varphi,\max}\left(v_{j,t}^{\phi} - v_{j,t}^{\varphi} - K_v\right)
\end{aligned}
\tag{C2}
$$

1) KKT stationarity conditions

Base on the Lagrangian function L_{DSO} in (C2), the stationarity conditions can be obtained as (C3) by requiring that the gradient of L_{DSO} with respect to the decision variables is zero.

- *The gradient for* $P_{ij,t}^{\phi}$

$$\frac{\partial L_{\mathrm{DSO}}}{\partial P_{ij,t}^{\phi_1}} = \pi_{i,t}^{p,\phi_1} - \pi_{j,t}^{p,\phi_1} - 2\left(\tilde{\boldsymbol{r}}_{ij}^{\mathrm{A}_{11}}\theta_{j,t}^{v,\phi_1} + \tilde{\boldsymbol{r}}_{ij}^{\mathrm{A}_{21}}\theta_{j,t}^{v,\phi_2} + \tilde{\boldsymbol{r}}_{ij}^{\mathrm{A}_{31}}\theta_{j,t}^{v,\phi_3}\right) = 0, \forall(i,j)\in\Omega_{\mathrm{L}}, \forall t\in\Omega_{\mathrm{T}} \tag{C3a}$$

$$\frac{\partial L_{\mathrm{DSO}}}{\partial P_{ij,t}^{\phi_2}} = \pi_{i,t}^{p,\phi_2} - \pi_{j,t}^{p,\phi_2} - 2\left(\tilde{\boldsymbol{r}}_{ij}^{\mathrm{A}_{12}}\theta_{j,t}^{v,\phi_1} + \tilde{\boldsymbol{r}}_{ij}^{\mathrm{A}_{22}}\theta_{j,t}^{v,\phi_2} + \tilde{\boldsymbol{r}}_{ij}^{\mathrm{A}_{32}}\theta_{j,t}^{v,\phi_3}\right) = 0, \forall(i,j)\in\Omega_{\mathrm{L}}, \forall t\in\Omega_{\mathrm{T}} \tag{C3b}$$

$$\frac{\partial L_{\mathrm{DSO}}}{\partial P_{ij,t}^{\phi_3}} = \pi_{i,t}^{p,\phi_3} - \pi_{j,t}^{p,\phi_3} - 2\left(\tilde{\boldsymbol{r}}_{ij}^{\mathrm{A}_{13}}\theta_{j,t}^{v,\phi_1} + \tilde{\boldsymbol{r}}_{ij}^{\mathrm{A}_{23}}\theta_{j,t}^{v,\phi_2} + \tilde{\boldsymbol{r}}_{ij}^{\mathrm{A}_{33}}\theta_{j,t}^{v,\phi_3}\right) = 0, \forall(i,j)\in\Omega_{\mathrm{L}}, \forall t\in\Omega_{\mathrm{T}} \tag{C3c}$$

where, taking the superscript A_{12} as an example, it represents the element in the first row and second column of the corresponding matrix.

- *The gradient for* $Q_{ij,t}^{\phi}$

$$\frac{\partial L_{\mathrm{DSO}}}{\partial Q_{ij,t}^{\phi_1}} = \pi_{i,t}^{q,\phi_1} - \pi_{j,t}^{q,\phi_1} - 2\left(\tilde{\boldsymbol{x}}_{ij}^{\mathrm{A}_{11}}\theta_{j,t}^{v,\phi_1} + \tilde{\boldsymbol{x}}_{ij}^{\mathrm{A}_{21}}\theta_{j,t}^{v,\phi_2} + \tilde{\boldsymbol{x}}_{ij}^{\mathrm{A}_{31}}\theta_{j,t}^{v,\phi_3}\right) = 0, \forall(i,j)\in\Omega_{\mathrm{L}}, \forall t\in\Omega_{\mathrm{T}} \tag{C4a}$$

$$\frac{\partial L_{\mathrm{DSO}}}{\partial Q_{ij,t}^{\phi_2}} = \pi_{i,t}^{q,\phi_2} - \pi_{j,t}^{q,\phi_2} - 2\left(\tilde{\boldsymbol{x}}_{ij}^{\mathrm{A}_{12}}\theta_{j,t}^{v,\phi_1} + \tilde{\boldsymbol{x}}_{ij}^{\mathrm{A}_{22}}\theta_{j,t}^{v,\phi_2} + \tilde{\boldsymbol{x}}_{ij}^{\mathrm{A}_{32}}\theta_{j,t}^{v,\phi_3}\right) = 0, \forall(i,j)\in\Omega_{\mathrm{L}}, \forall t\in\Omega_{\mathrm{T}} \tag{C4b}$$

$$\frac{\partial L_{\mathrm{DSO}}}{\partial Q_{ij,t}^{\phi_3}} = \pi_{i,t}^{q,\phi_3} - \pi_{j,t}^{q,\phi_3} - 2\left(\tilde{\boldsymbol{x}}_{ij}^{\mathrm{A}_{13}}\theta_{j,t}^{v,\phi_1} + \tilde{\boldsymbol{x}}_{ij}^{\mathrm{A}_{23}}\theta_{j,t}^{v,\phi_2} + \tilde{\boldsymbol{x}}_{ij}^{\mathrm{A}_{33}}\theta_{j,t}^{v,\phi_3}\right) = 0, \forall (i,j) \in \Omega_{\mathrm{L}}, \forall t \in \Omega_{\mathrm{T}} \tag{C4c}$$

- *The gradient for $v_{j,t}^{\phi}$*

For the root node of PDN, it has

$$\frac{\partial L_{\mathrm{DSO}}}{\partial v_{1,t}^{\phi}} = \theta_{2,t}^{v,\phi} - \theta_{1,t}^{v,\phi} = 0, \forall t \in \Omega_{\mathrm{T}}, \forall \phi \in \Phi \tag{C5}$$

For the non-root and non-terminal nodes in PDN, it has

$$\frac{\partial L_{\mathrm{DSO}}}{\partial v_{j,t}^{\phi_1}} = -\theta_{j,t}^{v,\phi_1} + \sum_{k \in C_j} \theta_{k,t}^{v,\phi_1} - \gamma_{j,t}^{v,\phi_1,\min} + \gamma_{j,t}^{v,\phi_1,\max} - \gamma_{j,t}^{\phi_1\phi_2,\min} + \gamma_{j,t}^{\phi_1\phi_2,\max} + \gamma_{j,t}^{\phi_3\phi_1,\min} - \gamma_{j,t}^{\phi_3\phi_1,\max} = 0, \tag{C6a}$$
$$\forall j \in \Omega_{\mathrm{N}}, \forall t \in \Omega_{\mathrm{T}}$$

$$\frac{\partial L_{\mathrm{DSO}}}{\partial v_{j,t}^{\phi_2}} = -\theta_{j,t}^{v,\phi_2} + \sum_{k \in C_j} \theta_{k,t}^{v,\phi_2} - \gamma_{j,t}^{v,\phi_2,\min} + \gamma_{j,t}^{v,\phi_2,\max} + \gamma_{j,t}^{\phi_1\phi_2,\min} - \gamma_{j,t}^{\phi_1\phi_2,\max} - \gamma_{j,t}^{\phi_2\phi_3,\min} + \gamma_{j,t}^{\phi_2\phi_3,\max} = 0, \tag{C6b}$$
$$\forall j \in \Omega_{\mathrm{N}}, \forall t \in \Omega_{\mathrm{T}}$$

$$\frac{\partial L_{\mathrm{DSO}}}{\partial v_{j,t}^{\phi_3}} = -\theta_{j,t}^{v,\phi_3} + \sum_{k \in C_j} \theta_{k,t}^{v,\phi_3} - \gamma_{j,t}^{v,\phi_3,\min} + \gamma_{j,t}^{v,\phi_3,\max} + \gamma_{j,t}^{\phi_2\phi_3,\min} - \gamma_{j,t}^{\phi_2\phi_3,\max} - \gamma_{j,t}^{\phi_3\phi_1,\min} + \gamma_{j,t}^{\phi_3\phi_1,\max} = 0, \tag{C6c}$$
$$\forall j \in \Omega_{\mathrm{N}}, \forall t \in \Omega_{\mathrm{T}}$$

For the terminal nodes in PDN, it has

$$\frac{\partial L_{\mathrm{DSO}}}{\partial v_{j,t}^{\phi_1}} = -\theta_{j,t}^{v,\phi_1} - \gamma_{j,t}^{v,\phi_1,\min} + \gamma_{j,t}^{v,\phi_1,\max} - \gamma_{j,t}^{\phi_1\phi_2,\min} + \gamma_{j,t}^{\phi_1\phi_2,\max} + \gamma_{j,t}^{\phi_3\phi_1,\min} - \gamma_{j,t}^{\phi_3\phi_1,\max} = 0, \forall j \in \Omega_{\mathrm{N}}, \forall t \in \Omega_{\mathrm{T}} \tag{C7a}$$

$$\frac{\partial L_{\mathrm{DSO}}}{\partial v_{j,t}^{\phi_2}} = -\theta_{j,t}^{v,\phi_2} - \gamma_{j,t}^{v,\phi_2,\min} + \gamma_{j,t}^{v,\phi_2,\max} + \gamma_{j,t}^{\phi_1\phi_2,\min} - \gamma_{j,t}^{\phi_1\phi_2,\max} - \gamma_{j,t}^{\phi_2\phi_3,\min} + \gamma_{j,t}^{\phi_2\phi_3,\max} = 0, \forall j \in \Omega_{\mathrm{N}}, \forall t \in \Omega_{\mathrm{T}} \tag{C7b}$$

$$\frac{\partial L_{\mathrm{DSO}}}{\partial v_{j,t}^{\phi_3}} = -\theta_{j,t}^{v,\phi_3} - \gamma_{j,t}^{v,\phi_3,\min} + \gamma_{j,t}^{v,\phi_3,\max} + \gamma_{j,t}^{\phi_2\phi_3,\min} - \gamma_{j,t}^{\phi_2\phi_3,\max} - \gamma_{j,t}^{\phi_3\phi_1,\min} + \gamma_{j,t}^{\phi_3\phi_1,\max} = 0, \forall j \in \Omega_{\mathrm{N}}, \forall t \in \Omega_{\mathrm{T}} \tag{C7c}$$

- *The gradient for $P_{\mathrm{sub},t}^{\phi}$*

$$\frac{\partial L_{\mathrm{DSO}}}{\partial P_{\mathrm{sub},t}^{\phi}} = \lambda_t^{\phi} - \pi_{1,t}^{p,\phi} - \gamma_{1,t}^{p,\phi,\min} + \gamma_{1,t}^{p,\phi,\max} = 0, \forall t \in \Omega_{\mathrm{T}}, \forall \phi \in \Phi \tag{C8}$$

- *The gradient for $Q_{\mathrm{sub},t}^{\phi}$*

$$\frac{\partial L_{\mathrm{DSO}}}{\partial Q_{\mathrm{sub},t}^{\phi}} = -\pi_{1,t}^{q,\phi} - \gamma_{1,t}^{q,\phi,\min} + \gamma_{1,t}^{q,\phi,\max} = 0, \forall t \in \Omega_{\mathrm{T}}, \forall \phi \in \Phi \tag{C9}$$

- *The gradient for $P_{j,t}^{\mathrm{MT},\phi}$ and $Q_{j,t}^{\mathrm{MT},\phi}$*

$$\frac{\partial L_{\mathrm{DSO}}}{\partial P_{j,t}^{\mathrm{MT},\phi}} = c_j^{\mathrm{MT}} - \pi_{j,t}^{p,\phi} - \gamma_{j,t}^{\mathrm{MT},p,\phi,\min} + \gamma_{j,t}^{\mathrm{MT},p,\phi,\max} - \gamma_{j,t}^{\mathrm{MT},q,\phi,\max} \tan(\arccos(\varphi^{\mathrm{pf}})) = 0, \forall j \in \Omega_{\mathrm{MT}}, \forall t \in \Omega_{\mathrm{T}}, \forall \phi \in \Phi \tag{C10a}$$

$$\frac{\partial L_{\mathrm{DSO}}}{\partial Q_{j,t}^{\mathrm{MT},\phi}} = -\pi_{j,t}^{q,\phi} - \gamma_{j,t}^{\mathrm{MT},q,\phi,\min} + \gamma_{j,t}^{\mathrm{MT},q,\phi,\max} = 0, \forall j \in \Omega_{\mathrm{MT}}, \forall t \in \Omega_{\mathrm{T}}, \forall \phi \in \Phi \tag{C10b}$$

2) Dual feasibility and complementary slackness conditions

$$0 \le \gamma_{j,t}^{v,\phi,\min} \perp v_{j,t}^{\phi} - \left(V_j^{\min,\phi}\right)^2 \ge 0, \forall j \in \Omega_{\mathrm{N}}, \forall t \in \Omega_{\mathrm{T}}, \forall \phi \in \Phi \tag{C11a}$$

$$0 \le \gamma_{j,t}^{v,\phi,\max} \perp \left(V_j^{\max,\phi}\right)^2 - v_{j,t}^{\phi} \ge 0, \forall j \in \Omega_{\mathrm{N}}, \forall t \in \Omega_{\mathrm{T}}, \forall \phi \in \Phi \tag{C11b}$$

$$0 \le \gamma_{1,t}^{p,\phi,\min} \perp P_{\mathrm{sub},t}^{\phi} + P_{\max} \ge 0, \forall t \in \Omega_{\mathrm{T}}, \forall \phi \in \Phi \tag{C11c}$$

$$0 \le \gamma_{1,t}^{p,\phi,\max} \perp P_{\max} - P_{\mathrm{sub},t}^{\phi} \ge 0, \forall t \in \Omega_{\mathrm{T}}, \forall \phi \in \Phi \tag{C11d}$$

$$0 \le \gamma_{1,t}^{q,\phi,\min} \perp Q_{\text{sub},t}^{\phi} + Q_{\max} \ge 0, \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C11e}$$

$$0 \le \gamma_{1,t}^{q,\phi,\max} \perp Q_{\max} - Q_{\text{sub},t}^{\phi} \ge 0, \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C11f}$$

$$0 \le \gamma_{j,t}^{\text{MT},p,\phi,\min} \perp P_{j,t}^{\text{MT},\phi} \ge 0, \forall j \in \Omega_{\text{MT}}, \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C11g}$$

$$0 \le \gamma_{j,t}^{\text{MT},p,\phi\max} \perp P_{j}^{\text{MT},\phi,\max} - P_{j,t}^{\text{MT},\phi} \ge 0, \forall j \in \Omega_{\text{MT}}, \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C11h}$$

$$0 \le \gamma_{j,t}^{\text{MT},q,\phi,\min} \perp Q_{j,t}^{\text{MT},\phi} \ge 0, \forall j \in \Omega_{\text{MT}}, \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C11i}$$

$$0 \le \gamma_{j,t}^{\text{MT},q,\phi\max} \perp P_{j,t}^{\text{MT},\phi} \tan(\arccos(\varphi^{\text{pf}})) - Q_{j,t}^{\text{MT},\phi} \ge 0, \forall j \in \Omega_{\text{MT}}, \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C11j}$$

$$0 \le \gamma_{j,t}^{\phi\varphi,\min} \perp v_{j,t}^{\phi} - v_{j,t}^{\varphi} + K_v \ge 0, \forall j \in \Omega_{\text{N}}, \forall t \in \Omega_{\text{T}}, \forall \phi, \varphi \in \Phi, \phi \ne \varphi \tag{C11k}$$

$$0 \le \gamma_{j,t}^{\phi\varphi,\max} \perp K_v - v_{j,t}^{\phi} + v_{j,t}^{\varphi} \ge 0, \forall j \in \Omega_{\text{N}}, \forall t \in \Omega_{\text{T}}, \forall \phi, \varphi \in \Phi, \phi \ne \varphi \tag{C11l}$$

The bilinear terms in (C12) can be relaxed by using Big-M method, presented as

$$0 \le \gamma_{j,t}^{v,\phi,\min} \le \text{M} Z_{j,t}^{v,\phi,\min}, 0 \le v_{j,t}^{\phi} - \left(V_{j}^{\min,\phi}\right)^2 \le \text{M}\left(1 - Z_{j,t}^{v,\phi,\min}\right), \forall j \in \Omega_{\text{N}}, \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C12a}$$

$$0 \le \gamma_{j,t}^{v,\phi,\max} \le \text{M} Z_{j,t}^{v,\phi,\max}, 0 \le \left(V_{j}^{\max,\phi}\right)^2 - v_{j,t}^{\phi} \le \text{M}\left(1 - Z_{j,t}^{v,\phi,\max}\right), \forall j \in \Omega_{\text{N}}, \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C12b}$$

$$0 \le \gamma_{1,t}^{p,\phi,\min} \le \text{M} Z_{1,t}^{p,\phi,\min}, 0 \le P_{\text{sub},t}^{\phi} + P_{\max} \le \text{M}\left(1 - Z_{1,t}^{p,\phi,\min}\right), \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C12c}$$

$$0 \le \gamma_{1,t}^{p,\phi,\max} \le \text{M} Z_{1,t}^{p,\phi,\max}, 0 \le P_{\max} - P_{\text{sub},t}^{\phi} \le \text{M}\left(1 - Z_{1,t}^{p,\phi,\max}\right), \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C12d}$$

$$0 \le \gamma_{1,t}^{q,\phi,\min} \le \text{M} Z_{1,t}^{q,\phi,\min}, 0 \le Q_{\text{sub},t}^{\phi} + Q_{\max} \le \text{M}\left(1 - Z_{1,t}^{q,\phi,\min}\right), \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C12e}$$

$$0 \le \gamma_{1,t}^{q,\phi,\max} \le \text{M} Z_{1,t}^{q,\phi,\max}, 0 \le Q_{\max} - Q_{\text{sub},t}^{\phi} \le \text{M}\left(1 - Z_{1,t}^{q,\phi,\max}\right), \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C12f}$$

$$0 \le \gamma_{j,t}^{\text{MT},p,\phi,\min} \le \text{M} Z_{j,t}^{\text{MT},p,\phi,\min}, 0 \le P_{j,t}^{\text{MT},\phi} \le \text{M}\left(1 - Z_{j,t}^{\text{MT},p,\phi,\min}\right), \forall j \in \Omega_{\text{N}}, \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C12g}$$

$$0 \le \gamma_{j,t}^{\text{MT},p,\phi,\max} \le \text{M} Z_{j,t}^{\text{MT},p,\phi,\max}, \quad 0 \le P_{j}^{\text{MT},\phi,\max} - P_{j,t}^{\text{MT},\phi} \le \text{M}\left(1 - Z_{j,t}^{\text{MT},p,\phi,\max}\right), \forall j \in \Omega_{\text{N}}, \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C12h}$$

$$0 \le \gamma_{j,t}^{\text{MT},q,\phi,\min} \le \text{M} Z_{j,t}^{\text{MT},q,\phi,\min}, 0 \le Q_{j,t}^{\text{MT},\phi} \le \text{M}\left(1 - Z_{j,t}^{\text{MT},q,\phi,\min}\right), \forall j \in \Omega_{\text{N}}, \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \tag{C12i}$$

$$\begin{aligned} 0 \le \gamma_{j,t}^{\text{MT},q,\phi,\max} \le \text{M} Z_{j,t}^{\text{MT},q,\phi,\max}, \quad 0 \le P_{j,t}^{\text{MT},\phi} \tan(\arccos(\varphi^{\text{pf}})) - Q_{j,t}^{\text{MT},\phi} \le \text{M}\left(1 - Z_{j,t}^{\text{MT},q,\phi,\max}\right), \\ \forall j \in \Omega_{\text{N}}, \forall t \in \Omega_{\text{T}}, \forall \phi \in \Phi \end{aligned} \tag{C12j}$$

$$0 \le \gamma_{j,t}^{\phi\varphi,\min} \le \text{M} Z_{j,t}^{\phi\varphi,\min}, 0 \le v_{j,t}^{\phi} - v_{j,t}^{\varphi} + K_v \le \text{M}\left(1 - Z_{j,t}^{\phi\varphi,\min}\right), \forall j \in \Omega_{\text{N}}, \forall t \in \Omega_{\text{T}}, \forall \phi, \varphi \in \Phi, \phi \ne \varphi \tag{C12k}$$

$$0 \le \gamma_{j,t}^{\phi\varphi,\max} \le \text{M} Z_{j,t}^{\phi\varphi,\max}, 0 \le K_v - v_{j,t}^{\phi} + v_{j,t}^{\varphi} \le \text{M}\left(1 - Z_{j,t}^{\phi\varphi,\max}\right), \forall j \in \Omega_{\text{N}}, \forall t \in \Omega_{\text{T}}, \forall \phi, \varphi \in \Phi, \phi \ne \varphi \tag{C12l}$$

where, M is a large constant and all the Z-type variables are 0-1 variables.

3) Primal feasibility conditions

The primal feasible conditions have been detailed in (C1).

Dual objective function reformulation of the DSO problem

By substituting the KKT stationarity conditions (C3)-(C10) into the Lagrangian function L_{DSO} in (C2), L_{DSO} can be reformulated as the objective function of the dual problem in (C1), expressed as (C13).

$$
\begin{aligned}
&L_{\mathrm{DSO}}\left(\begin{array}{l}\pi_{j,t}^{p,\phi},\pi_{j,t}^{q,\phi},\theta_{j,t}^{v,\phi},\gamma_{j,t}^{v,\phi,\min},\gamma_{j,t}^{v,\phi,\max},\gamma_{1,t}^{p,\phi,\min},\gamma_{1,t}^{p,\phi,\max},\gamma_{1,t}^{q,\phi,\min},\gamma_{1,t}^{q,\phi,\max},\gamma_{j,t}^{\mathrm{MT},p,\phi,\min},\gamma_{j,t}^{\mathrm{MT},p,\phi,\max},\\ \gamma_{j,t}^{\mathrm{MT},q,\phi,\min},\gamma_{j,t}^{\mathrm{MT},q,\phi,\max},\gamma_{j,t}^{\phi\varphi,\min},\gamma_{j,t}^{\phi\varphi,\max}\end{array}\right)\\
&=\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left(\pi_{j,t}^{p,\phi}P_{j,t}^{\mathrm{D},\phi}+\pi_{j,t}^{q,\phi}Q_{j,t}^{\mathrm{D},\phi}+\pi_{j,t}^{p,\phi}P_{j,t}^{\mathrm{Hub},\phi}\right)\\
&+\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left[\gamma_{j,t}^{v,\phi,\min}\left(V_j^{\min,\phi}\right)^2-\gamma_{j,t}^{v,\phi,\max}\left(V_j^{\max,\phi}\right)^2+\theta_{1,t}^{v,\phi}\left(V^{ref}\right)^2\right]\\
&-\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left[P_{\max}\left(\gamma_{1,t}^{p,\phi,\min}+\gamma_{1,t}^{p,\phi,\max}\right)+Q_{\max}\left(\gamma_{1,t}^{q,\phi,\min}+\gamma_{1,t}^{q,\phi,\max}\right)\right]\\
&-\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\substack{(\phi,\varphi)\in\Phi\\ \phi\neq\varphi}}K_v\left(\gamma_{j,t}^{\phi\varphi,\min}+\gamma_{j,t}^{\phi\varphi,\max}\right)-\sum_{j\in\Omega_{\mathrm{MT}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\gamma_{j,t}^{\mathrm{MT},p,\phi,\max}P_j^{\mathrm{MT},\phi,\max}
\end{aligned}
\tag{C13}
$$

According to the strong duality theorem, the objective function of the primal problem is equal to that of the dual problem, that is

$$
\begin{aligned}
&\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\lambda_t^{\phi}P_{\mathrm{sub},t}^{\phi}+\sum_{j\in\Omega_{\mathrm{MT}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}c_j^{\mathrm{MT}}P_{j,t}^{\mathrm{MT},\phi}\\
&=\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left(\pi_{j,t}^{p,\phi}P_{j,t}^{\mathrm{D},\phi}+\pi_{j,t}^{q,\phi}Q_{j,t}^{\mathrm{D},\phi}+\pi_{j,t}^{p,\phi}P_{j,t}^{\mathrm{Hub},\phi}\right)\\
&+\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left[\gamma_{j,t}^{v,\phi,\min}\left(V_j^{\min,\phi}\right)^2-\gamma_{j,t}^{v,\phi,\max}\left(V_j^{\max,\phi}\right)^2+\theta_{1,t}^{v,\phi}\left(V^{ref}\right)^2\right]\\
&-\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left[P_{\max}\left(\gamma_{1,t}^{p,\phi,\min}+\gamma_{1,t}^{p,\phi,\max}\right)+Q_{\max}\left(\gamma_{1,t}^{q,\phi,\min}+\gamma_{1,t}^{q,\phi,\max}\right)\right]\\
&-\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\substack{(\phi,\varphi)\in\Phi\\ \phi\neq\varphi}}K_v\left(\gamma_{j,t}^{\phi\varphi,\min}+\gamma_{j,t}^{\phi\varphi,\max}\right)-\sum_{j\in\Omega_{\mathrm{MT}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\gamma_{j,t}^{\mathrm{MT},p,\phi,\max}P_j^{\mathrm{MT},\phi,\max}
\end{aligned}
\tag{C14}
$$

The equivalent MILP problem

Based on the equation $\lambda_{j,t}^{\mathrm{DLMP},\phi}=\pi_{j,t}^{p,\phi}$, we can derive an equivalent form of the objective function in the SCHO[1] subproblem by transposing the terms in (C14), thereby effectively eliminating the bilinear terms from the original objective function.

$$
\begin{aligned}
&-\sum_{j\in\Omega_{\mathrm{N_{Hub}}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\lambda_{j,t}^{\mathrm{DLMP},\phi}P_{j,t}^{\mathrm{Hub},\phi}\\
&=-\sum_{j\in\Omega_{\mathrm{N_{Hub}}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\pi_{j,t}^{p,\phi}P_{j,t}^{\mathrm{Hub},\phi}\\
&=\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left(\pi_{j,t}^{p,\phi}P_{j,t}^{\mathrm{D},\phi}+\pi_{j,t}^{q,\phi}Q_{j,t}^{\mathrm{D},\phi}\right)\\
&-\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\lambda_t^{\phi}P_{\mathrm{sub},t}^{\phi}-\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\substack{(\phi,\varphi)\in\Phi\\ \phi\neq\varphi}}K_v\left(\gamma_{j,t}^{\phi\varphi,\min}+\gamma_{j,t}^{\phi\varphi,\max}\right)\\
&-\sum_{j\in\Omega_{\mathrm{MT}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left(c_j^{\mathrm{MT}}P_{j,t}^{\mathrm{MT},\phi}+\gamma_{j,t}^{\mathrm{MT},p,\phi,\max}P_j^{\mathrm{MT},\max}\right)\\
&+\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left[\gamma_{j,t}^{v,\phi,\min}\left(V_j^{\min,\phi}\right)^2-\gamma_{j,t}^{v,\phi,\max}\left(V_j^{\max,\phi}\right)^2+\theta_{1,t}^{v,\phi}\left(V^{ref}\right)^2\right]\\
&-\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left[P_{\max}\left(\gamma_{1,t}^{p,\phi,\min}+\gamma_{1,t}^{p,\phi,\max}\right)+Q_{\max}\left(\gamma_{1,t}^{q,\phi,\min}+\gamma_{1,t}^{q,\phi,\max}\right)\right]
\end{aligned}
\tag{C15}
$$

Thus, the original objective function in the SCHO[1] subproblem can be replaced by (C16)

$$
\begin{aligned}
&\max -\sum_{j\in\Omega_{\mathrm{N_{Hub}}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\lambda_{j,t}^{\mathrm{DLMP},\phi}P_{j,t}^{\mathrm{Hub},\phi}\\
&=\max\left\{\begin{array}{l}
\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left(\pi_{j,t}^{p,\phi}P_{j,t}^{\mathrm{D},\phi}+\pi_{j,t}^{q,\phi}Q_{j,t}^{\mathrm{D},\phi}\right)\\
-\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\lambda_{t}^{\phi}P_{\mathrm{sub},t}^{\phi}-\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\substack{(\phi,\varphi)\in\Phi\\ \phi\neq\varphi}}K_{v}\left(\gamma_{j,t}^{\phi\varphi,\min}+\gamma_{j,t}^{\phi\varphi,\max}\right)\\
-\sum_{j\in\Omega_{\mathrm{MT}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left(c_{j}^{\mathrm{MT}}P_{j,t}^{\mathrm{MT},\phi}+\gamma_{j,t}^{\mathrm{MT},p,\phi,\max}P_{j}^{\mathrm{MT},\max}\right)\\
+\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left[\gamma_{j,t}^{v,\phi,\min}\left(V_{j}^{\min,\phi}\right)^{2}-\gamma_{j,t}^{v,\phi,\max}\left(V_{j}^{\max,\phi}\right)^{2}+\theta_{1,t}^{v,\phi}\left(V^{ref}\right)^{2}\right]\\
-\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left[P_{\max}\left(\gamma_{1,t}^{p,\phi,\min}+\gamma_{1,t}^{p,\phi,\max}\right)+Q_{\max}\left(\gamma_{1,t}^{q,\phi,\min}+\gamma_{1,t}^{q,\phi,\max}\right)\right]
\end{array}\right\}
\end{aligned}
\tag{C16}
$$

The constraints of the SCHO[1] subproblem include (6)-(8). The KKT conditions of the DSO problem consist of the primal feasibility constraints (2) and (3), the KKT stationarity conditions (C3)-(C10), as well as the dual feasibility and relaxed complementary slackness conditions (C12). In summary, the equivalent MILP problem can be formulated as

$$
\begin{aligned}
&\max\left\{\begin{array}{l}
\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left(\pi_{j,t}^{p,\phi}P_{j,t}^{\mathrm{D},\phi}+\pi_{j,t}^{q,\phi}Q_{j,t}^{\mathrm{D},\phi}\right)\\
-\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\lambda_{t}^{\phi}P_{\mathrm{sub},t}^{\phi}-\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\substack{(\phi,\varphi)\in\Phi\\ \phi\neq\varphi}}K_{v}\left(\gamma_{j,t}^{\phi\varphi,\min}+\gamma_{j,t}^{\phi\varphi,\max}\right)\\
-\sum_{j\in\Omega_{\mathrm{MT}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left(c_{j}^{\mathrm{MT}}P_{j,t}^{\mathrm{MT},\phi}+\gamma_{j,t}^{\mathrm{MT},p,\phi,\max}P_{j}^{\mathrm{MT},\max}\right)\\
+\sum_{j\in\Omega_{\mathrm{N}}}\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left[\gamma_{j,t}^{v,\phi,\min}\left(V_{j}^{\min,\phi}\right)^{2}-\gamma_{j,t}^{v,\phi,\max}\left(V_{j}^{\max,\phi}\right)^{2}+\theta_{1,t}^{v,\phi}\left(V^{ref}\right)^{2}\right]\\
-\sum_{t\in\Omega_{\mathrm{T}}}\sum_{\phi\in\Phi}\left[P_{\max}\left(\gamma_{1,t}^{p,\phi,\min}+\gamma_{1,t}^{p,\phi,\max}\right)+Q_{\max}\left(\gamma_{1,t}^{q,\phi,\min}+\gamma_{1,t}^{q,\phi,\max}\right)\right]
\end{array}\right\}\\
&s.t.\quad \begin{array}{l}\text{cons-DSO}:(2)-(3),(C3)-(C10),(C12)\\ \text{cons-SCHO}^{1}:(6)-(8)\end{array}
\end{aligned}
\tag{C18}
$$